

5

Nonparametric methods

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5.1 Introduction

In some studies, the questions of scientific interest can be addressed without specifying a parametric model for the data. A venerable, and generic, example is design-based inference for randomised experiments, which dates back at least to the early work of R. A. Fisher at Rothamsted; see, for example, Fisher, R.A. (1925, 1935).

In this chapter, we discuss methods for analysing spatial point pattern data that are not tied to particular parametric families of models. The material builds directly on the discussion of second-order properties in Section 4.6.

5.2 Estimating weighted integrals of the second-order intensity

Recall that one definition of the K -function of a stationary process is

$$K(t) = 2\pi\lambda^{-2} \int_0^t \lambda_2(s) s ds \quad (5.1)$$

where λ and $\lambda_2(s)$ are the intensity and second-order intensity, respectively. It turns out that several non-parametric inference problems for point process data can be solved by estimating a weighted integral,

$$K_\phi(t) = 2\pi\lambda^{-2} \int_0^t \phi(s)\lambda_2(s)sd s, \quad (5.2)$$

for suitably defined, problem-specific functions $\phi(s)$. Before considering a specific application, we give the general result, from Berman and Diggle (1989).

It is convenient to re-express (5.2) as

$$K_\phi(t) = 2\pi\lambda^{-2}H(t)$$

where

$$H(t) = \int_0^t \phi(s)\lambda_2(s)sd s. \quad (5.3)$$

Using integration by parts to evaluate (5.3), and substituting from (5.1), we obtain

$$H(t) = \lambda^2(2\pi)^{-1} \left(K(t)\phi(t) - \int_0^t K(s)\phi'(s)ds \right) \quad (5.4)$$

Estimation of $H(t)$ is now straightforward, because $\phi(\cdot)$ is a known function and we can substitute existing estimators for λ and for $K(t)$ into (5.4). In practice, the integration on the right hand side of (5.4) must be carried out numerically, but this is usually straightforward and numerically more stable than would have been the case for the direct numerical evaluation of (5.3). Given a sufficiently large data-set, it would also be feasible to substitute an estimate of $\lambda_2(t)$ into (5.3) and apply numerical integration to the right hand side of (5.3), but for the reasons discussed in Section 4.2, we prefer the estimator $\hat{H}(t)$ based on (5.4).

5.3 Nonparametric estimation of a spatially varying intensity

Suppose that the available data are a partial realisation of a Cox process, and that we wish to estimate the realisation of $\Lambda(x)$, the underlying intensity process. A simple and intuitively sensible estimator would consist of counting, for each location x , the number of events of the process within a distance h of x and scaling by πh^2 , the area of a disc of radius h . In practice, we shall need to adjust this simple estimator to allow for edge-effects when x is close to the boundary of the study region, but we can ignore this complication for the time being. How should we choose the value of h ? One way is to consider the mean square error of the resulting estimator.

Diggle (1985b) derived the mean square error of the estimator on the assumption that the underlying process is a stationary, isotropic Cox process. If the driving intensity of a Cox process has expectation μ and covariance function $\gamma(u)$, then the Cox process itself has intensity $\lambda = \mu$ and second-order intensity $\lambda_2(u) = \gamma(u) + \mu^2$. Let $N(x, h)$ denote the number of events of the Cox process within distance h of the point x . Then, temporarily ignoring edge-effects, the non-parametric estimator of the realised value of $\Lambda(x)$ described above can be written as

$$\tilde{\lambda}(x) = N(x, h)/(\pi h^2). \quad (5.5)$$

We now consider the mean square error of $\tilde{\lambda}(x)$,

$$MSE(h) = E[\{\tilde{\lambda}(x) - \Lambda(x)\}^2],$$

where the expectation is with respect to the distribution of the Cox process, i.e. with respect both to $\Lambda(\cdot)$ and to the points of the process conditional on $\Lambda(\cdot)$. Stationarity implies that $MSE(h)$ does not depend on x . Taking $x = 0$ and using a standard conditioning argument, we have that

$$\begin{aligned} MSE(h) &= E_{\Lambda}[E_N\{N/(\pi h^2) - \Lambda(0)\}^2] \\ &= E_{\Lambda}[\text{Var}_N\{N/(\pi h^2)\} + \{E_N[N/(\pi h^2)] - \Lambda(0)\}^2], \end{aligned} \quad (5.6)$$

where $N = N(0, h)$. Conditional on $\Lambda(\cdot)$, the count $N(0, h)$ follows a Poisson distribution with both mean and variance equal to $\int \Lambda(x)dx$, where the integration is over the disc with centre 0 and radius h . Hence, (5.6) becomes

$$\begin{aligned} MSE(h) &= E_{\Lambda} \left[\int \Lambda(x)dx/(\pi h^2)^2 + \int \int \Lambda(x)\Lambda(y)dydx \right. \\ &\quad \left. - 2 \int \Lambda(x)\Lambda(0)dx + \lambda(0)^2 \right] \\ &= \lambda/(\pi h^2) + \int \int \lambda_2(\|x - y\|)dydx - 2 \int \lambda_2(\|x\|)dx + \lambda_2(0). \end{aligned}$$

Now, use the fact that $\int \lambda_2(\|x\|)dx = \lambda^2 K(h)$ to give

$$MSE(h) = \lambda_2(0) + \lambda\{1 - 2\lambda K(h)\}/(\pi h^2) + (\pi h^2)^{-2} \int \int \lambda_2(\|x - y\|)dydx. \quad (5.7)$$

The first term on the right hand side of (5.7) does not depend on h , and it follows that the value of h which minimises $MSE(h)$ also minimises

$$M(h) = \lambda\{1 - 2\lambda K(h)\}/(\pi h^2) + (\pi h^2)^{-2} \int \int \lambda_2(\|x - y\|)dydx. \quad (5.8)$$

The first of the two terms on the right hand side of (5.8) is an explicit function of $K(h)$, and can be estimated by substituting the standard estimator $\hat{K}(\cdot)$.

The double integral in the second term can be converted to a single integral of the form (5.2) using polar coordinates, and can therefore be estimated as described in Section 5.1.

We now consider how to deal with edge-effects in the estimator $\tilde{\lambda}(x)$. Several methods of edge-correction are available. The one used by Diggle (1985b) in the one-dimensional case and extended to the two-dimensional case by Berman and Diggle (1989) replaces the denominator πh^2 in (5.5) by the area of intersection of the relevant disc with the study region. Hence, if the data are observed on a region A and $B(x, h)$ denotes the disc with centre x and radius h , the edge-corrected estimator is

$$\hat{\lambda}(x) = N(x, h) / |A \cap B(x, h)| \quad (5.9)$$

A further, and in this case largely cosmetic, refinement can be made by interpreting the estimator as a kernel estimator (Silverman, 1981). Define a kernel function $k(u)$ to be any radially symmetric, bivariate pdf (expressed in polar coordinates); thus, $k(u) \geq 0$ for all $u \geq 0$ and

$$2\pi \int_0^\infty k(u)u du = 1.$$

Then, $k_h(u) = h^{-2}k(u/h)$ is also a radially symmetric pdf for any $h > 0$, and a kernel estimator of a bivariate pdf $f(x)$, based on data $x_1, \dots, x_n$, takes the form

$$\hat{f}(x) = n^{-1} \sum_{i=1}^n k_h(x - x_i).$$

The estimator $\tilde{\lambda}(x)$ can now be seen as a special case of the kernel estimator, with kernel function

$$k(u) = \begin{cases} (\pi u^2)^{-1} & : 0 \leq u \leq 1 \\ 0 & : u > 1. \end{cases} \quad (5.10)$$

Note, however, that the expressions given here for $\tilde{\lambda}(x)$ and $\hat{f}(x)$ differ by a factor of n , because the intensity is a mean number of events per unit area and, unlike the pdf, does not integrate to 1.

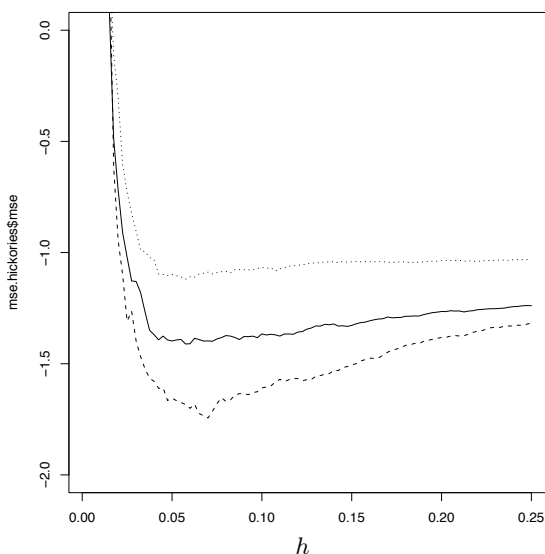
Viewed in this light, the edge-corrected estimator $\hat{\lambda}(x)$ can be written as

$$\hat{\lambda}(x) = \sum_{i=1}^n k_h(u) / \int_A k_h(||x||) dx$$

Most applications of kernel density estimation would use a smoother kernel function than (5.10), for example the quartic

$$k(u) = \begin{cases} 3\pi^{-1}(1 - u^2)^2 & : 0 \leq u \leq 1 \\ 0 & : u > 1. \end{cases} \quad (5.11)$$

In this case one way to choose the value of h is to calibrate (5.11) against (5.10)

**FIGURE 5.1**

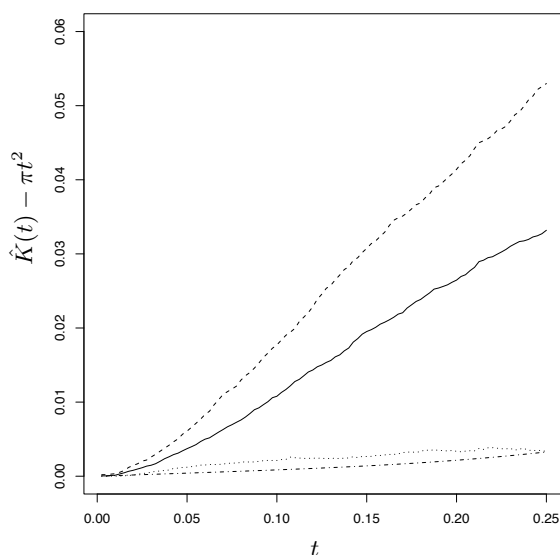
Estimates of the mean square error, $\hat{M}(h)$ of a non-parametric intensity estimator applied to each of the three major species groupings in Lansing Woods. — — — — — : hickories; — — — : maples; : oaks.

by equating second moments. For the bivariate random variable, (X, Y) say, with pdf (5.11) the expectation of $U^2 = X^2 + Y^2$ is $h^2/4$, whereas with the circular uniform pdf (5.10) the corresponding expectation is $h^2/2$. Hence, if h_0 is the optimum bandwidth for a uniform kernel according to (5.8), the calibrated bandwidth for the quartic kernel is $h = h_0\sqrt{2}$.

In the original setting of non-parametric probability density estimation, Silverman (1981) points out that to obtain an estimate $\hat{f}(x)$ with good properties, the precise choice of kernel function is relatively unimportant by comparison with the choice of h . This is also generally true in the present context. An exception is when the underlying intensity surface is highly variable, in particular including large empty regions, and interest is in the ratio of intensities between two data-sets; this question arises naturally, for example in the application of point process methods to case-control studies in epidemiology, as we shall discuss in Chapter 9.

5.3.1 Estimating spatially varying intensities for the Lansing Woods data

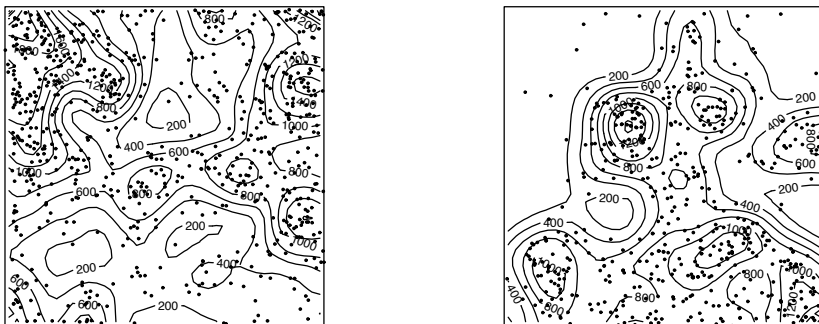
Figure 5.1 shows estimates of the function $M(h)$ obtained by applying (5.8) to each of the three major species groupings of the Lansing Woods data shown

**FIGURE 5.2**

Estimates $\hat{K}(t) - \pi t^2$ for each of the three major species groupings in Lansing Woods. — : hickories; — — : maples; : oaks. Pointwise values of two standard errors assuming an underlying homogeneous Poisson process are shown as - · - · -

originally as Figure 2.11. We suggest that a plot of $\hat{M}(h)$ should be used as a guide to the choice of h , rather than as an automatic procedure. In this respect, we emphasise two particular features. Firstly, because $\hat{M}(h)$ is an empirical quantity and therefore subject to sampling variation, multiple local minima are to be expected in the neighbourhood of the optimal value of h where $M(h)$ is relatively flat. Secondly, and as illustrated here by the plot for the oaks, near-monotonicity of $\hat{M}(h)$ is a useful indication that there is little or no evidence in the data for spatial variation in intensity. When the underlying Cox process reduces to a homogeneous Poisson process, for which $\Lambda(x)$ is constant and $\gamma(u) = 0$ for all u , the theoretical form of $MSE(h)$ reduces to $MSE(h) = (\pi h^2)^{-1}$, which is monotone decreasing in h . The estimates shown in Figure 5.1 have been scaled so that they in fact estimate $\{MSE(h) + \lambda_2(0)\}/\lambda^2$. This implies that for a homogeneous Poisson process, $\hat{M}(h) \rightarrow -1$ as $h \rightarrow \infty$.

The estimated K -function can also be used directly to confirm that the data are indeed spatially aggregated, and that a Cox process is a reasonable working model, before constructing an estimate of the realised intensity surface $\Lambda(x)$. Figure 5.2 shows $\hat{K}(t) - \pi t^2$ for each of the three species. The pointwise two-standard-error limits for a homogeneous Poisson process with the same intensity as the oaks are also shown as a dot-dashed line. This indicates that the second-order properties of the oaks are close to, albeit sig-

**FIGURE 5.3**

Kernel estimates of $\lambda(x)$ for the hickories (left-hand panel) and maples (right-hand panel) in Lansing Woods, using a quartic kernel with bandwidth $h = 0.1$.

nificantly different from, those of a homogeneous Poisson process whereas the hickories and maples are unequivocally spatially aggregated.

We therefore interpret Figure 5.1 as follows. For both the hickories and the maples, the minimum estimated mean square error is substantially less than -1 , indicating spatial variation in intensity, whereas for the oaks the estimate $\hat{M}(h)$ is approximately constant for h greater than about 0.03 and never substantially less than -1 . For the maples, there is a reasonably well-defined minimum of $\hat{M}(h)$ around $h \approx 0.07$, whereas the trace of $\hat{M}(h)$ for the hickories is minimised at $h \approx 0.06$, but is then rather flat until about $h \approx 0.09$. Our preliminary conclusion is that there is substantial spatial variation in the intensity of hickories and maples in Lansing Woods, whereas the oaks, although departing significantly from complete spatial randomness on the evidence of Figure 5.2, show approximately constant intensity. In general, we would consider using non-parametric estimation of spatially varying intensity $\lambda(x)$ only when, as is the case here, the data-set contains several hundred events or more, and when there is unequivocal evidence of substantial variation in $\lambda(x)$.

The two panels of Figure 5.3 show the estimates of $\lambda(x)$ obtained for the hickories and maples, using the quartic kernel (5.11) with $h = 0.07\sqrt{2} \approx 0.1$ in both cases to make the estimates directly comparable.

Notice that the two estimated surfaces $\hat{\lambda}(x)$ are essentially complementary to each other, in the sense that the intensity of hickories is high where the intensity of maples is low, and *vice versa*, suggesting that the two species occupy distinct ecological niches. We obtained a qualitatively similar result in Section (7.1.1) using a parametric model for $\lambda(x)$. However, with the selected bandwidth $h = 0.1$, the non-parametric estimates capture more of the variation in the data, the contour lines showing considerably more complicated behaviour

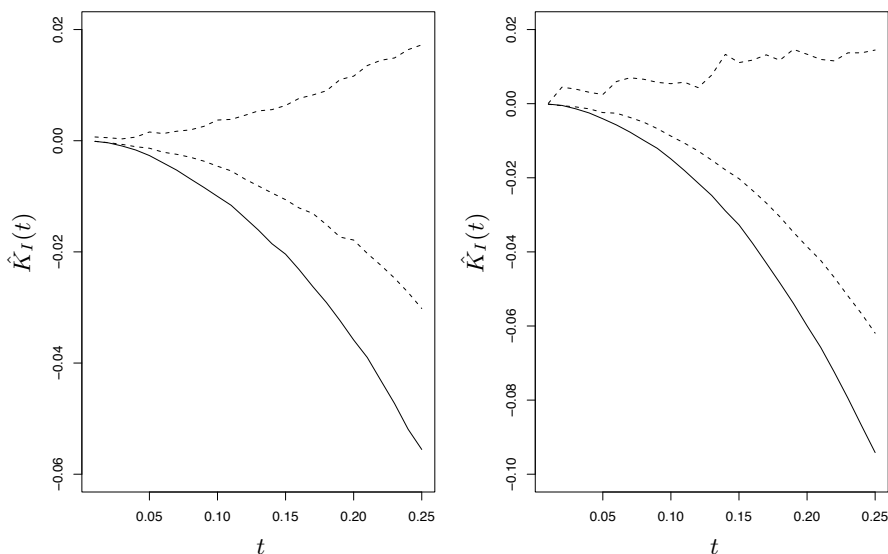
than those of the parametric model. As noted earlier, the oaks display a pattern which is close to Poisson, implying in the present context a near-constant intensity, or an ability to exploit either of the two niches favoured by the hickories and by the maples. Of course, at a sufficiently small scale, we would expect to find competitive interactions between near-neighbouring trees, in violation of the Cox process assumption. Note, in this context, that the average area per tree in the whole forest (including a small number of miscellaneous trees not considered in this analysis) is approximately 35 square metres.

In contrast to the results for the parametric analysis of these data that we shall present in Section 8.2, if we assume intensity-reweighted stationarity and estimate $K_I(t)$ by using our non-parametric estimates of $\lambda(x)$ in Figure 5.3, then for both the hickories and the maples $\hat{K}_I(t)$ is significantly *less* than πt^2 . Figure 5.4 shows, for each species, $\hat{K}_I(t)$ together with upper and lower envelopes from 19 simulations of an inhomogeneous Poisson process with intensity equal to the corresponding kernel estimate $\hat{\lambda}(x)$. The apparent regularity suggests small-scale inhibitory effects between neighbouring trees, but could also be a by-product of over-fitting a complicated surface $\hat{\lambda}(x)$.

The parametric analysis of these data that we shall discuss in Section 8.2.1 uses a log-quadratic model for the intensity, $\lambda(x)$. It could be argued that the nonparametric methodology described here is better able to describe subtle gradations in intensity over the study region. On the other hand, there is an attendant risk that a nonparametric smoothing method will find spurious features in the estimated intensity surface when used on sparse data. This underlines the importance of examining a plot of $\hat{M}(h)$ before computing the surface estimate $\hat{\lambda}(\cdot)$. A more important point relates to the theoretical requirement that $\lambda(x)$ should be bounded away from zero. In practice, this means that when the study-region A contain sub-regions where the intensity is close to zero, the estimator $\hat{K}_I(t)$ can be very sensitive to the choice of estimator for $\lambda(x)$.

5.4 Analysing replicated spatial point patterns

Nonparametric methods of inference are also appropriate when the data consist of replicate point patterns within a designed experiment, in which case inference can be based on the design rather than on an assumed stochastic model. For example, in Section 4.6 we considered the problem of estimating the sampling variance of an estimator for $K(t)$. There, we distinguished between situations in which the underlying process was or was not assumed to be a homogeneous Poisson process. In the second of these cases, we considered two different ways in which we might proceed, leading to model-based and design-based inference, respectively. For data consisting of a single point pattern, the design-based method used a form of pseudo-replication by divid-

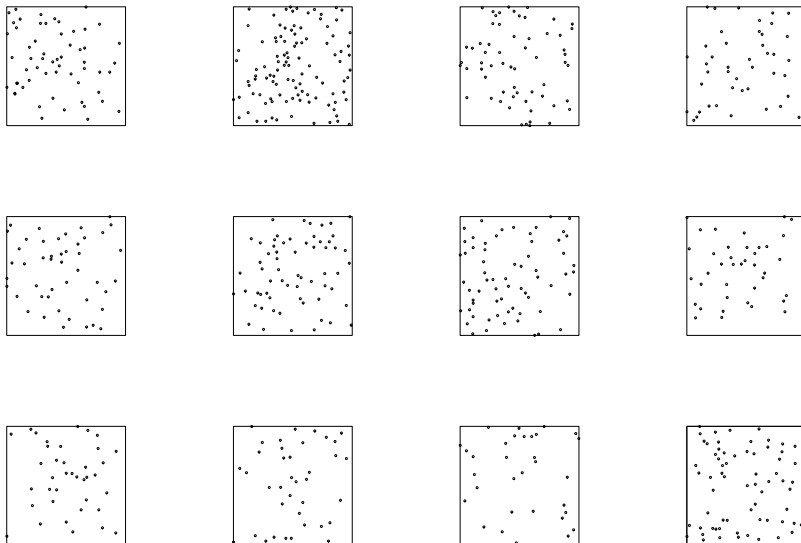
**FIGURE 5.4**

Estimates of $K_I(t)$ for the hickories (left-hand panel) and maples (right-hand panel) in Lansing Woods. — — — — : data; — — — upper and lower envelopes from 19 simulations of the inhomogeneous Poisson process with intensity equal to the corresponding kernel estimate $\hat{\lambda}(\cdot)$.

ing the study region A into sub-regions. We now consider how to analyse data with genuine replication.

Genuine replication is obtained when an observed point pattern is the result of an experiment that can be repeated under identical conditions, thus producing a sequence of patterns which, by design, are exchangeable. It is then natural to analyse the resulting data using the design-based approach to inference.

As a motivating example, we consider the data shown in Figure 5.5. These consist of 12 point patterns, each of which identifies the locations of pyramidal neurons within a microscopic section of brain tissue taken from an area of the cingulate cortex (area 24, layer 2) of a human subject post-mortem (Diggle, Lange and Benes, 1991). The 12 subjects are presumed to be a random sample of normal brains, and it is of interest to quantify the typical spatial arrangement of pyramidal neurons, for later comparison with samples from abnormal subjects. It would be difficult to justify a stationary process model for the pattern presented by a single individual. Nevertheless, we can use the estimated K -function as a general summary measure of spatial aggregation, and the between-subject variation in estimated K -functions as the basis for inference.

**FIGURE 5.5**

Locations of pyramidal neurons in brain tissue sections from 12 normal subjects.

5.4.1 Estimating the K -function from replicated data

Suppose that the data consist of r point patterns, each observed on a region A . Denote the i th of these by $\mathcal{X}_i = \{x_{i1}, \dots, x_{in_i}\}$. When the patterns are strict replicates of an underlying process, the corresponding estimates of the K -function are identically distributed and a reasonable overall estimate can be obtained by simple averaging, as was done for the estimator $\hat{K}(t)$ discussed briefly in Section (4.6.1). Because $K(t)$ is itself defined as a ratio, $K(t) = E(t)/\lambda$, a better strategy might be to pool separately estimates of λ and of $E(t) = \lambda K(t)$. This leads to $\hat{\lambda} = \sum n_i / (r|A|)$, which is the maximum likelihood estimator when the underlying process is a homogeneous Poisson process, and $\hat{E}(t) = \sum \hat{E}_i(t) / r$, where $\hat{E}_i(t)$ is the estimate of $E(t)$ obtained from the i th replicate. The resulting estimator for $K(t)$ is

$$\hat{K}(t) = \sum_{i=1}^r n_i \hat{K}_i(t) / \sum_{i=1}^r n_i, \quad (5.12)$$

a weighted average of the individual estimates $\hat{K}_i(t)$.

The K -function is defined so as not to depend on the underlying intensity of events. We can therefore also construct a pooled estimated without assuming a common intensity across all replicates; this presumes that the hypothesis of a common K -function and varying intensity between replicates is scientifically

plausible, as would be the case if the “replicates” were differentially thinned versions of a common underlying process. However, in this case we would again argue that the weighted average (5.12) is an appropriate estimator, since the dominant term in the variance of $\hat{K}_i(t)$ is of order n_i^{-1} . This also holds if the regions on which the separate patterns are observed differ in size. The distinction between assuming a common intensity λ or pattern-specific intensities λ_i might be expected to make a material difference. However, in the latter case, the argument leading to the estimator (5.12) holds directly, whereas in the former, it follows by using the estimators

$$\tilde{E}(t) = \sum_{i=1}^r |A_i| \hat{E}_i(t) / \sum_{i=1}^r |A_i|$$

and $\tilde{\lambda} = \sum_{i=1}^r n_i / \sum_{i=1}^r |A_i|$ together with the fact that $\hat{E}_i(t) = n_i \hat{K}_i(t) / |A_i|$.

For a design-based assessment of the sampling variance of $\hat{K}(t)$, we use a simple method based on the bootstrap (Efron and Tibshirani, 1993). Define *residual K-functions*,

$$R_i(t) = n_i^{0.5} \{ \hat{K}_i(t) - \hat{K}(t) \} : i = 1, \dots, r \quad (5.13)$$

To a first approximation, the $R_i(t)$ are exchangeable under the assumption that the underlying processes may differ in their intensities, but are otherwise identical. We then construct a bootstrap sample of K -functions as

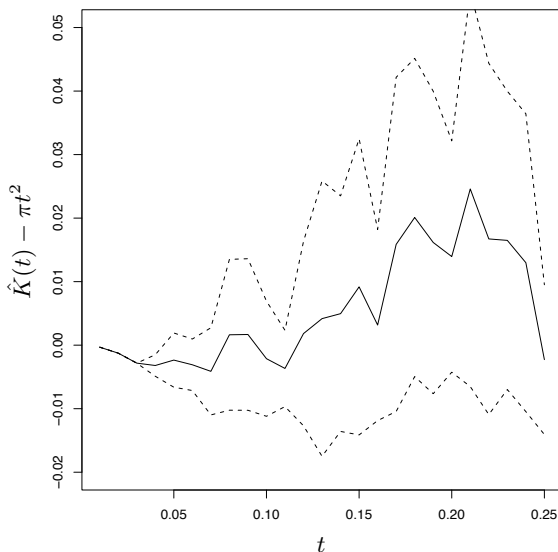
$$K_i^*(t) = \hat{K}(t) + n_i^{-0.5} R_i^*(t) : i = 1, \dots, r,$$

where the $R_i^*(\cdot)$ are sampled at random with replacement from the set $\{R_1(\cdot), \dots, R_r(\cdot)\}$, and compute a re-sampled $\hat{K}(t)$ as

$$\hat{K}^*(t) = \sum_{i=1}^r n_i \hat{K}_i^*(t) / \sum_{i=1}^r n_i.$$

Repeating this whole procedure, say s times, we then take the sample variance of the s values of $\hat{K}^*(t)$ as the bootstrap approximation to the sampling variance of $\hat{K}(t)$. Note that, because the whole of each residual K -function is re-sampled, the procedure gives a bootstrap estimate of the variance matrix of the vector of values of $\hat{K}(t)$ for the complete set of values of t under consideration, if required.

Figure 5.6 shows the estimate $\hat{K}(t) - \pi t^2$, as defined at (5.12), for the data in Figure 5.5, together with plus and minus two bootstrap standard error limits, computed from $s = 1000$ re-samples. Using a larger number of re-samples did not materially change the bootstrap variance estimates. All of the original estimated K -functions, and hence necessarily all of their bootstrap re-sampled counterparts, have $\hat{K}(t) = 0$ for small distances $t < 0.03$, approximately, indicating a small-scale inhibitory effect. At larger distances, the rapidly increasing width of the bootstrap standard error limits renders non-significant any further departure from an underlying $K(t) = \pi t^2$.

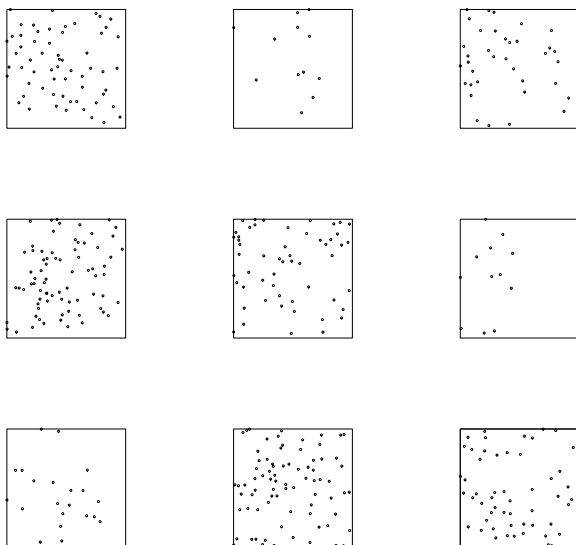
**FIGURE 5.6**

Pooled estimate $\hat{K}(t) - \pi t^2$ from 12 control subjects, with bootstrapped plus and minus two pointwise standard error limits

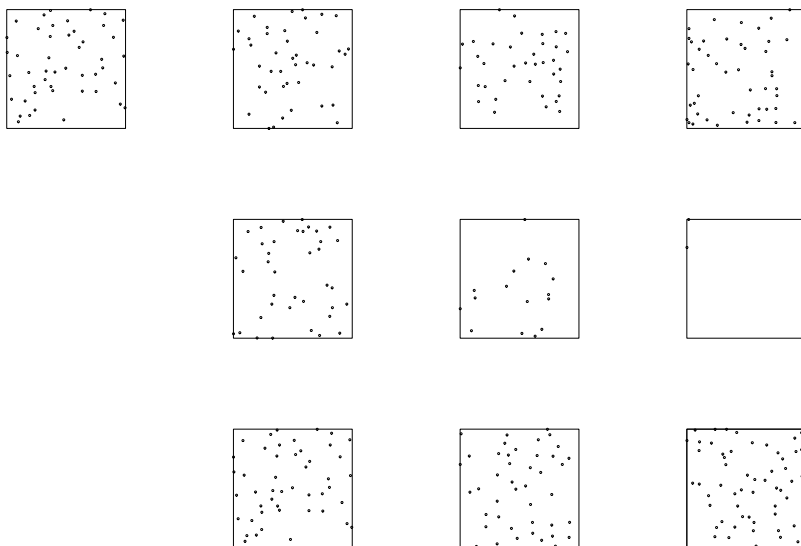
5.4.2 Between-group comparisons in designed experiments

We now extend our motivating example to include data from two additional experimental groups, consisting of post-mortem samples from subjects previously diagnosed as schizo-affective or schizophrenic. These data are shown in Figures 5.7 and 5.8. One of the patterns from the schizophrenic group has only two pyramidal neurons and we shall ignore it in the subsequent analysis.

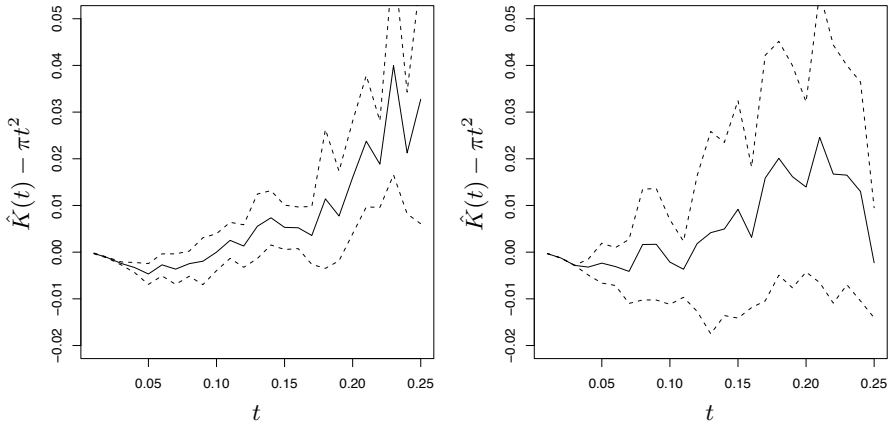
Preliminary analysis of the counts n_i , using a log-linear Poisson regression model, suggested that the intensity of events varied significantly between groups. The observed mean numbers of cells per pattern in the three groups were 54.6 for the controls, 45.1 for the schizo-affectives and 37.4 (excluding the pattern with only two cells) for the schizophrenics. The likelihood ratio statistic to test for equality of the three underlying population means within the Poisson log-linear model was 33.3 on 2 degrees of freedom. There is no formal justification for assuming Poisson counts in this context, but unless the individual patterns are spatially aggregated, a Poisson approximation should be conservative (McCullagh and Nelder, 1989). Recall that the estimator $\hat{K}(t)$ remains valid when the underlying intensities vary between replicates, and therefore also when the intensities vary between groups. The two panels of Figure 8.9 show the resulting estimates of $K(t) - \pi t^2$ for the schizo-affectives and for the schizophrenics, together with pointwise limits constructed as plus and minus two bootstrap standard errors, and using a common vertical scale

**FIGURE 5.7**

Locations of pyramidal neurons in brain tissue sections from 9 schizo-affective subjects.

**FIGURE 5.8**

Locations of pyramidal neurons in brain tissue sections from 10 schizophrenic subjects.

**FIGURE 5.9**

Pooled estimates $\hat{K}(t) - \pi t^2$ from 9 schizo-affective subjects (left-hand panel) and from 9 schizophrenic subjects (right-hand panel), with bootstrapped plus and minus two pointwise standard error limits.

for all three groups. Visual inspection of Figures 5.6 and 5.9 suggests that all three groups show small-scale inhibitory behaviour, as is to be expected in most micro-anatomical applications where the events represent reference locations for finite-sized objects. However, the groups appear to differ in their larger-scale behaviour; specifically, the schizo-affectives alone show apparently significant spatial aggregation. We now consider how to convert this visual assessment into a formal inference.

Let n_{ij} denote the number of events for the j th subject within the i th experimental group, and $\hat{K}_{ij}(t)$ the corresponding estimated K -function. Define $\hat{K}_i(t)$ to be the estimate (5.12) calculated from the r_i subjects in the i th group, and $\hat{K}_0(t)$ a weighted average of the $\hat{K}_i(t)$ with weights proportional to the total numbers of events, $n_i = \sum_{j=1}^{r_i} n_{ij}$, in the three groups. Now let $K_i(t)$ denote the expectation of $\hat{K}_{ij}(t)$ under repeated sampling. The null hypothesis is that this is the same in all three groups, hence $K_i(t) = K(t)$. Then, the $\hat{K}_i(t)$ estimate the corresponding $K_i(t)$, whilst $\hat{K}_0(t)$ estimates $K(t)$ under the null hypothesis that $K_i(t) = K(t)$ for all i .

To test the hypothesis that $K_i(t) = K(t)$, a statistic loosely analogous to the between treatment sum of squares in a classical analysis of variance is

$$BTSS = \sum_{i=1}^3 n_i \int_0^{t_0} w(t) \{ \hat{K}_i(t) - \hat{K}_0(t) \}^2 dt. \quad (5.14)$$

The null sampling distribution of T is intractable, but for a design-based

inference we can again use a re-sampling method. For the j th subject within the i th group, define the *residual K -function*,

$$R_{ij}(t) = n_{ij}^{0.5} \{ \hat{K}_{ij}(t) - \hat{K}_i(t) \} \quad (5.15)$$

The $R_{ij}(t)$ are, to a first approximation, exchangeable under the null or alternative hypotheses. Hence, if $R_{ij}^*(t) : j = 1, \dots, r_i : i = 1, 2, 3$ are obtained from the original $R_{ij}(t)$ by re-sampling, we can construct a set of re-sampled K -functions under the null hypothesis as

$$\hat{K}_{ij}^*(t) = \hat{K}_0(t) + n_{ij}^{-0.5} R_{ij}^*(t) \quad (5.16)$$

For an approximate test, we then compare the observed value, $BTSS_1$ say, with values $BTSS_k : k = 2, \dots, s$ from independent sets of re-sampled $K_{ij}^*(t)$.

We applied the bootstrap procedure with $t_0 = 0.25$, $w(t) = t^{-2}$ and 1000 re-samples. Note that in this context, each re-sample is a complete set of 30 estimated K -functions in three groups, but generated under the null hypothesis that the three underlying group-mean K -functions are equal. The resulting bootstrap p -value is 0.253, giving no reason formally to reject the null hypothesis.

Diggle, Lange and Benes (1991) used an unweighted version of the test statistic (5.14) in conjunction with a square root transformation of $\hat{K}(t)$. They also used random permutations of the residual K -functions, i.e. re-sampling without replacement, rather than the bootstrap re-sampling with replacement implemented here.

5.5 Parametric or nonparametric methods?

We have argued that for the application described in Section 5.4, parametric modelling assumptions would be hard to justify, and a nonparametric, design-based approach therefore seems natural. Of course, in this context an estimated K -function is just one of many possible *ad hoc* summary statistics that could be calculated from each pattern. It is a reasonable choice when there is scientific interest in the degree of spatial aggregation or regularity in the component patterns and in the extent to which this varies between groups of subjects. For more specific alternative hypotheses, other summary statistics may be preferable. Note also that Bell and Grunwald (2004) have analysed the same set of data using parametric methods. They fit pairwise interaction point process models using maximum pseudo-likelihood and accommodate differences between replicates by treating model parameters as random effects.

An immediate benefit of using likelihood-based methods when a suitable parametric model can be identified is that *ad hoc* methods of inference are no

longer necessary. Also, information from independent replicates can be combined objectively, by adding the corresponding log-likelihood contributions. The corresponding cost is the reliance on additional assumptions, i.e. the correctness of the assumed model. An intermediate strategy is to use likelihood-based methods to estimate model parameters as summary statistics for each replicate, but to continue to use the randomisation distribution induced by the study design as the basis for inference. Diggle, Mateu and Clough (2000) report some empirical comparisons of the parametric and nonparametric approaches. Their results confirm, as expected, that the parametric approach is more powerful when its underlying assumptions are satisfied, but correspondingly less robust to departures from the assumptions.